

Chapter 1

Most Stable Z and Mass Parabolas

“For constant A , the mass formula represents a parabola of M vs Z .”

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1.1 Atomic mass from the SEMF

Lecture 4 constructed the binding energy $B(A, Z)$ and explained how its five terms shape the B/A curve. Stability against β decay is decided not by B alone but by the *atomic* mass $M(A, Z)$: at fixed mass number A , the isobar with the lowest mass cannot lower its energy by converting a neutron into a proton (or vice versa). Writing M as a function of Z at fixed A produces the mass parabolas of this lecture [6, 1, 14].

Key idea

From binding energy to atomic mass.

$$M(A, Z) c^2 = [Z m(^1\text{H}) + (A - Z) m_n] c^2 - B(A, Z).$$

β decay changes Z by ± 1 at fixed A , so it moves along an isobaric slice of this surface. The SEMF turns that slice into a parabola in Z .

Key idea

Which SEMF terms matter at fixed A ? When A is held constant, the volume term $a_V A$ and the surface term $a_S A^{2/3}$ are the same for every isobar and drop out of mass *differences*. What remains are:

- the free-nucleon mass piece, linear in Z through $Z m(^1\text{H}) + (A - Z) m_n$;
- Coulomb, quadratic in Z ;
- asymmetry, quadratic in Z through $(A - 2Z)^2$;
- pairing δ , constant on each even/odd Z ladder.

Lecture 4 fixed the bulk physics of B/A ; Lecture 5 extracts the Z dependence that controls β stability.

The SEMF binding energy is

$$B(A, Z) = a_V A - a_S A^{2/3} - a_C \frac{Z(Z-1)}{A^{1/3}} - a_a \frac{(N-Z)^2}{A} + \delta(A, Z), \quad (1.1)$$

with $N = A - Z$ and

$$\delta = \begin{cases} +a_P A^{-3/4} & \text{even-even,} \\ 0 & \text{odd } A, \\ -a_P A^{-3/4} & \text{odd-odd.} \end{cases} \quad (1.2)$$

The atomic mass energy is

$$M(A, Z) c^2 = [Z m(^1\text{H}) + (A - Z) m_n] c^2 - B(A, Z). \quad (1.3)$$

(Using $m(^1\text{H})$ rather than m_p folds the electron mass into the atomic ledger, which is what β energetics require.)

Substituting (1.1) into (1.3) makes the Z dependence explicit:

$$\begin{aligned} M(A, Z) c^2 = & [Z m(^1\text{H}) + (A - Z) m_n] c^2 \\ & - a_V A + a_S A^{2/3} + a_C \frac{Z(Z-1)}{A^{1/3}} + a_a \frac{(A - 2Z)^2}{A} - \delta. \end{aligned} \quad (1.4)$$

Volume and surface depend only on A . Pairing is constant on each even/odd lattice of Z . The free nucleon (atomic) masses contribute a term linear in Z . Coulomb and asymmetry contribute quadratic pieces in Z .

1.2 Why $M(A, Z)$ is a parabola in Z

Expand the Z -dependent pieces at fixed A . With $N - Z = A - 2Z$,

$$\begin{aligned} a_a \frac{(A - 2Z)^2}{A} &= \frac{a_a}{A} (A^2 - 4AZ + 4Z^2) \\ &= a_a A - 4a_a Z + \frac{4a_a}{A} Z^2, \end{aligned} \quad (1.5)$$

and

$$a_C \frac{Z(Z-1)}{A^{1/3}} = \frac{a_C}{A^{1/3}} Z^2 - \frac{a_C}{A^{1/3}} Z. \quad (1.6)$$

The free-nucleon piece contributes $[m(^1\text{H}) - m_n] c^2 Z$. Collecting powers of Z ,

$$M(A, Z) c^2 = \alpha(A) + \beta(A) Z + \gamma(A) Z^2 - \delta, \quad (1.7)$$

where

$$\beta(A) = [m(^1\text{H}) - m_n] c^2 - 4a_a - \frac{a_C}{A^{1/3}}, \quad (1.8)$$

$$\gamma(A) = \frac{a_C}{A^{1/3}} + \frac{4a_a}{A} > 0. \quad (1.9)$$

(The linear asymmetry coefficient is $-4a_a$, not $+4a_a/A$: the latter belongs only to the quadratic piece.) The constant $\alpha(A)$ collects volume, surface, the $a_a A$ piece, and the $A m_n c^2$ offset; it does not affect which Z is most stable. Because $\gamma > 0$, $M(Z)$ is a parabola opening *upward*: there is a unique minimum in Z , and nuclei far from that minimum have higher mass and can β decay toward it [6, 1].

Differentiating the quadratic form recovers the same stationarity condition used below: $\partial M/\partial Z = \beta + 2\gamma Z$, and setting that to zero with (1.8)–(1.9) reproduces (1.10).

Geometrically, Coulomb and asymmetry both contribute positive curvature ($\propto Z^2$). Coulomb raises the mass of proton-rich isobars; asymmetry raises the mass when $|N - Z|$ is large. The minimum is the compromise charge where those penalties balance, moderated by the small linear term $\beta(A)Z$ from nucleon mass differences and the linear piece of the Coulomb expansion.

The most stable isobar is the integer Z nearest the minimum of that parabola.

Key idea

At fixed A , Coulomb repulsion and asymmetry energy together make the mass a quadratic in Z . Minimising mass maximises stability against β decay.

1.3 Most stable Z : differentiation of the mass formula

1.3.1 Setting the derivative to zero

Ignore pairing for the moment (exact for odd A ; for even A the pairing shifts two parallel parabolas but does not move their common $Z_{\min}$ at leading order). Start from (1.4) and differentiate with respect to Z at fixed A . Only the Z -dependent pieces contribute:

$$\frac{\partial}{\partial Z} [M(A, Z) c^2] = \underbrace{[m(^1\text{H}) - m_n] c^2}_{\text{free-nucleon slope}} + \underbrace{a_C \frac{(2Z - 1)}{A^{1/3}}}_{\text{Coulomb slope}} - \underbrace{a_a \frac{4(A - 2Z)}{A}}_{\text{asymmetry slope}}. \quad (1.10)$$

The asymmetry derivative uses $\partial(A - 2Z)^2/\partial Z = -4(A - 2Z)$. Setting the derivative to zero gives

$$a_C \frac{(2Z - 1)}{A^{1/3}} - [m_n - m(^1\text{H})] c^2 = 4a_a \frac{(A - 2Z)}{A}. \quad (1.11)$$

Equivalently, the Coulomb slope equals the neutron–hydrogen mass drive *plus* the asymmetry slope. A plus sign before the mass difference on the left would not reproduce Equation (1.13). Collecting terms that multiply Z ,

$$Z \left(\frac{2a_C}{A^{1/3}} + \frac{8a_a}{A} \right) = [m_n - m(^1\text{H})] c^2 + \frac{a_C}{A^{1/3}} + 4a_a, \quad (1.12)$$

and therefore the most stable charge

$$Z_{\min} = \frac{(m_n - m(^1\text{H})) c^2 + a_C A^{-1/3} + 4a_a}{2a_C A^{-1/3} + 8a_a/A}. \quad (1.13)$$

This is the exact stationary condition for the SEMF mass parabola [6, 1]. The integer Z closest to $Z_{\min}$ is the smooth-SEMF candidate; pairing, shell structure, and evaluated atomic masses

decide the actual beta-stable member.

1.3.2 Numerical size of the free-mass term

From atomic mass units,

$$m_n - m(^1\text{H}) = 1.008665 - 1.007825 = 0.000840 \text{ u}, \quad (1.14)$$

so

$$[m_n - m(^1\text{H})]c^2 \approx 0.000840 \times 931.5 \approx 0.782 \text{ MeV}. \quad (1.15)$$

With the usual coefficients $a_C = 0.72 \text{ MeV}$ and $a_a = 23 \text{ MeV}$, for $A \sim 100$ one has

$$\frac{a_C}{A^{1/3}} \approx 0.16 \text{ MeV}, \quad \frac{4a_a}{A} \approx 0.92 \text{ MeV},$$

while $4a_a = 92 \text{ MeV}$ dominates the numerator of (1.13). The free-mass difference is not negligible next to $a_C/A^{1/3}$, but both are small compared with the asymmetry scale [6].

1.3.3 Working formula used in the lecture

Neglecting the small free-mass and linear-Coulomb pieces relative to $4a_a$ in the numerator, and writing

$$Z_{\min} \approx \frac{4a_a}{2a_C A^{-1/3} + 8a_a/A} = \frac{A/2}{1 + \frac{1}{4}(a_C/a_a) A^{2/3}}, \quad (1.16)$$

one obtains with $a_C/a_a = 0.72/23 \approx 0.0313$

$$Z_{\min} \approx \frac{A}{2(1 + 0.0078 A^{2/3})}. \quad (1.17)$$

A closely related fit uses 0.0075 instead of 0.0078 in the denominator [14].

Key idea

Light vs heavy nuclei. For small A , $0.0078 A^{2/3} \ll 1$, so $Z_{\min} \approx A/2$ ($N \approx Z$): oxygen, carbon, calcium. For large A the Coulomb correction grows, $Z_{\min} < A/2$, and long-lived nuclei become neutron rich ($N/Z \sim 1.5$ for lead). Equation (1.17) gives $Z/A \approx 0.41$ for heavy species, matching observation [6, 1].

A	$Z_{\min}$ (approx.)	Nearest stable Z	N/Z
16	7.6	8 (^{16}O)	1.0
40	18.3	18 (^{40}Ar); ^{40}Ca also stable	1.22 at $Z = 18$
56	25.1	26 (^{56}Fe)	1.2
125	52.3	52 (^{125}Te)	1.4
208	81.6	82 (^{208}Pb)	1.5

Values of $Z_{\min}$ come from (1.17). At $A = 40$ the smooth formula places the minimum near argon, not calcium. ^{40}Ca is nevertheless stable because its doubly magic shell closure supplies binding

absent from the SEMF. The nearest integer to a smooth minimum is therefore a guide, not an evaluated-mass verdict.

1.4 Odd A : one parabola and β decay toward the minimum

For odd A , $\delta = 0$. There is a *single* mass parabola. Nuclei with Z far from $Z_{\min}$ have large mass differences to their neighbours and decay by $\beta^\pm$ or electron capture; nuclei near the minimum are stable or long-lived.

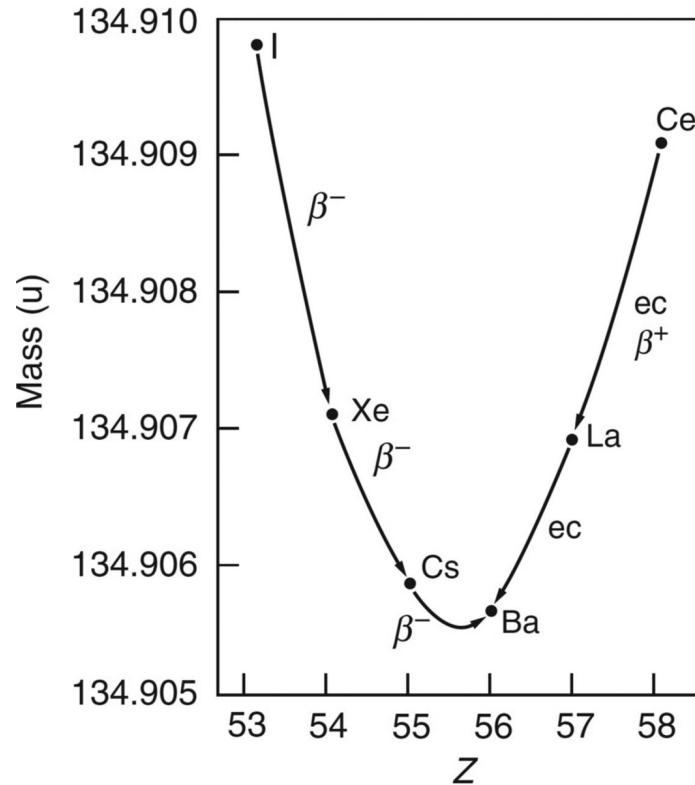


Figure 1.1: Mass parabola for odd $A = 135$ (original lecture figure). One minimum near $Z = 56$; one (or few) β -stable isobar(s). Decay energy increases as one moves away from the minimum [1, 6].

1.4.1 β energetics on the parabola

Along an odd- A isobar the available energy for a single β step is the atomic mass difference of neighbouring isobars. For β^- ($Z \rightarrow Z + 1$),

$$Q_{\beta^-} = [M(A, Z) - M(A, Z + 1)]c^2. \quad (1.18)$$

For β^+ ($Z \rightarrow Z - 1$),

$$Q_{\beta^+} = [M(A, Z) - M(A, Z - 1)]c^2 - 2m_e c^2 \quad (1.19)$$

(the extra $2m_e c^2$ appears when atomic masses are used carefully for positron emission). Electron capture has a slightly lower threshold than β^+ .

Because $M(Z)$ is parabolic, the mass difference between neighbours *grows* as one moves away from $Z_{\min}$. That is why β -decay Q values (and typically the decay rates) increase as one climbs the sides of the parabola [6, 1].

Key idea

Direction of β decay from geometry alone.

$Z < Z_{\min}$	neutron rich	β^- ($n \rightarrow p + e^- + \bar{\nu}_e$),
$Z > Z_{\min}$	proton rich	β^+ or electron capture,
$Z \approx Z_{\min}$	stable / long-lived	no allowed single β .

The SEMF fixes the *direction* and approximate Q ; Fermi theory fixes spectra and lifetimes.

1.5 Even A : pairing splits the mass surface into two parabolas

For even A , the pairing term shifts even–even nuclei *down* and odd–odd nuclei *up* by $|\delta| = a_P A^{-3/4}$ in the binding-energy convention of (1.1). Because $M c^2$ contains $-\delta$, the mass of an even–even isobar is lowered by $|\delta|$ and that of an odd–odd isobar is raised by $|\delta|$. The vertical separation between the two parabolas is therefore $2|\delta|$. The mass surface becomes two parallel parabolas sharing the same $Z_{\min}$ but offset in energy [6, 1, 14].

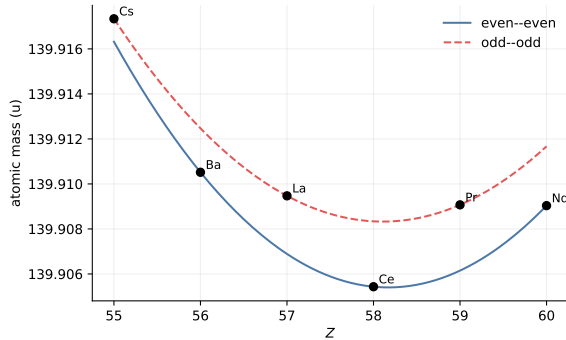


Figure 1.2: Even $A = 140$: pairing splits even–even (lower) and odd–odd (upper) chains. One β -stable even–even isobar near $Z = 58$.

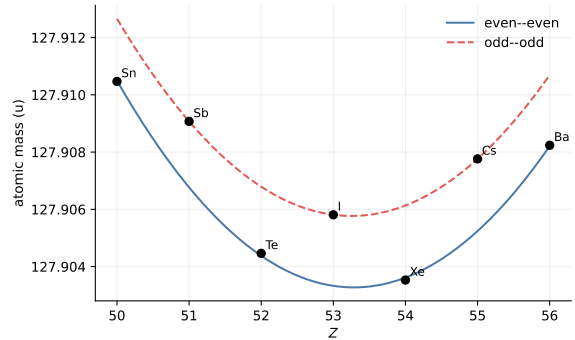


Figure 1.3: Even $A = 128$: two even–even isobars ($Z = 52, 54$) lie on the lower parabola; the odd–odd member can decay both ways. Some even–even chains are stable against *single* β but decay very slowly by double β (e.g. tellurium isotopes).

1.5.1 Two effects absent from odd- A chains

1. **Bidirectional β decay of odd–odd nuclei.** An odd–odd isobar near the minimum of the *upper* parabola can have both neighbours on the lower parabola at lower mass. Then both β^- and β^+ /EC are open. Classic example: ^{128}I can convert either a neutron or a proton to reach a more bound even–even neighbour [6, 1].
2. **Double β decay of even–even nuclei.** If single β from an even–even nucleus would land on a higher odd–odd isobar, that single step is energetically forbidden, yet a transition to a lower even–even neighbour two units of Z away can still be allowed. The nucleus may then

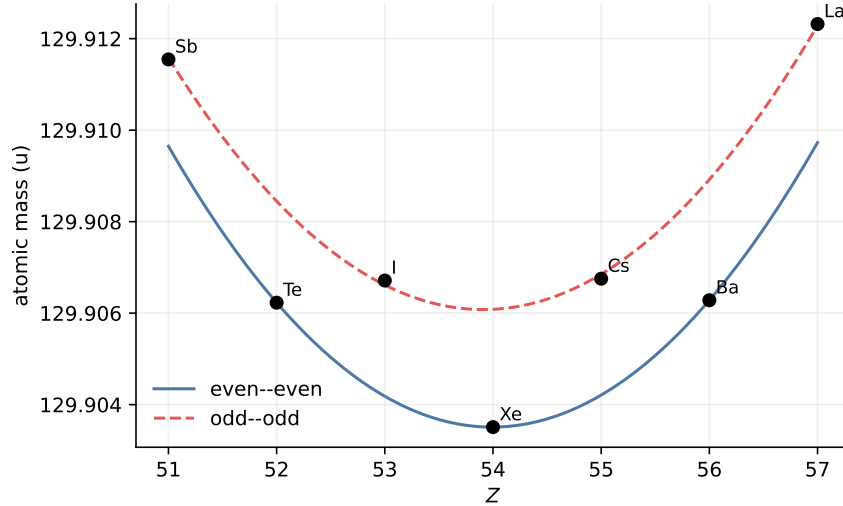


Figure 1.4: Even $A = 130$: three even–even nuclides ($Z = 52, 54, 56$) on the lower parabola can all be stable against single β decay relative to their odd–odd neighbours. Long-term evolution may still proceed by double β where $Q_{2\beta} > 0$.

decay only by **double beta decay** ($2\nu\beta\beta$, or the still-unseen $0\nu\beta\beta$). Candidates include ^{76}Ge , ^{128}Te , ^{130}Te , and ^{136}Xe , where single β is blocked by pairing but $Q_{2\beta} > 0$ [6, 1, 4]. Observationally, such nuclei can be extremely long lived: ^{128}Te has a $2\nu\beta\beta$ half-life of order 10^{24} yr.

Key idea

Pairing does not invent new forces; it rearranges which isobars are allowed to β decay. Odd A : one parabola, one valley. Even A : two parabolas, possible double- β and two-way decays.

1.5.2 Why light odd–odd nuclei can be stable

For small A the parabola is narrow (the coefficient $\gamma(A)$ is larger). The pairing offset can then place an odd–odd nucleus low enough to be stable against single β . The four stable odd–odd species ^2H , ^6Li , ^{10}B , ^{14}N live in that regime; heavier odd–odd nuclei almost always find an open β path to an even–even daughter [1, 6].

1.6 β Q values from the SEMF (worked structure)

Using atomic masses, the β^- energy release between neighbouring isobars is

$$Q_{\beta^-}(A, Z) = [M(A, Z) - M(A, Z + 1)]c^2 = [m_n - m(^1\text{H})]c^2 + B(A, Z + 1) - B(A, Z) \quad (1.20)$$

(up to the usual atomic-mass bookkeeping). Substituting the SEMF, volume and surface cancel at fixed A , and one is left with Coulomb, asymmetry, and pairing differences, exactly the terms that shape the parabola [3, 14].

Similarly for β^+ ,

$$Q_{\beta^+}(A, Z) = [M(A, Z) - M(A, Z - 1)]c^2 - 2m_e c^2, \quad (1.21)$$

so both Coulomb (fewer protons helps) and asymmetry (moving toward $N = Z$ or away from it) enter with opposite roles on the two sides of the valley.

Worked estimate: Q_{β^-} for ^{135}Cs

For odd $A = 135$, pairing vanishes and the SEMF gives a single parabola. With $a_C = 0.72$, $a_a = 23$ (MeV), $Z_{\min} \approx 56$ so the valley sits near barium ($Z = 56$). Consider neutron-rich ^{135}Cs ($Z = 55$):

$$Q_{\beta^-} = [M(135, 55) - M(135, 56)]c^2.$$

Volume and surface cancel at fixed A . A finite difference must use the *same* parent and daughter charge in every term. The Coulomb contribution to $M(Z) - M(Z + 1)$ is

$$a_C A^{-1/3} [Z(Z - 1) - (Z + 1)Z] = -a_C A^{-1/3} (2Z) = -0.72 \cdot \frac{110}{5.13} \approx -15.4 \text{ MeV}.$$

The asymmetry contribution to $M(Z) - M(Z + 1)$ is

$$a_a A^{-1} [(A - 2Z)^2 - (A - 2Z - 2)^2] = a_a A^{-1} [4(A - 2Z) - 4] \approx 23 \cdot \frac{4 \cdot 25 - 4}{135} \approx 16.36 \text{ MeV}.$$

Altogether

$$Q_{\beta^-} \approx 0.782 - 15.44 + 16.36 \approx 1.70 \text{ MeV}.$$

The evaluated atomic-mass difference gives only $Q_{\beta^-} \approx 0.269 \text{ MeV}$ [27]. This is not a contradiction: ^{135}Cs lies very close to the parabola minimum, where a sub-MeV shell residual is comparable to the entire finite difference. The SEMF captures the direction and rough scale; evaluated masses, not the smooth formula, determine whether a near-valley decay is open and its precision Q value. The positive sign confirms β^- decay of neutron-rich ^{135}Cs toward ^{135}Ba .

Use evaluated masses for decay decisions

The continuous SEMF vertex answers “where is the valley approximately?” It does not certify an integer nuclide as stable. For neighbouring isobars compute

$$Q_{\beta^-} = [M_{\text{at}}(A, Z) - M_{\text{at}}(A, Z + 1)]c^2$$

(and the corresponding β^+ or electron-capture expression) from an evaluated mass table such as AME2020. Pairing and shell corrections that are small in a total mass can dominate this near-cancellation [27].

Physics checklist for an isobar

1. Compute $Z_{\min}(A)$ from (1.17) (or (1.13)).
2. For odd A : one parabola; the integer Z nearest $Z_{\min}$ is stable; others β decay toward it with Q increasing up the sides.

3. For even A : draw two parabolas split by $\sim 2|\delta|$; identify even–even minima and any odd–odd bidirectional cases; mark possible double- β pairs.
4. Q_β follows from mass differences on that surface, with no weak matrix element required for the *sign* of the decay.

1.7 Connecting the valley of stability

Plotting the integer nearest to $Z_{\min}(A)$ for each A traces the valley of stability on the (N, Z) plane: $N \approx Z$ at low A , bending to $N > Z$ at high A . Lectures 4–5 therefore close the liquid-drop programme: binding energy and its Z landscape, including the β arrows that keep real nuclei near the valley [6, 1, 13].

1.8 Deeper physics: exact $Z_{\min}$ and isobar energetics

1.8.1 Complete mass formula in nuclear masses

The same parabola follows from writing the atomic mass with nuclear constituents,

$$M(A, Z) = (A - Z)m_n + Z(m_p + m_e) - \frac{a_V A}{c^2} + \frac{a_S A^{2/3}}{c^2} + \frac{a_C Z(Z - 1)}{c^2 A^{1/3}} + \frac{a_a (A - 2Z)^2}{c^2 A} - \frac{\delta}{c^2}, \quad (1.22)$$

which differs from (1.4) only by the usual $m(^1\text{H})$ versus $m_p + m_e$ bookkeeping. Setting $\partial M / \partial Z = 0$ yields

$$Z_{\min} = \frac{(m_n - m_p - m_e)c^2 + a_C A^{-1/3} + 4a_a}{2a_C A^{-1/3} + 8a_a/A}, \quad (1.23)$$

equivalent to (1.13). The free-mass difference $(m_n - m_p - m_e)c^2 \approx 0.78 \text{ MeV}$ is small next to $4a_a \sim 90 \text{ MeV}$; dropping it and the linear Coulomb piece recovers (1.17) [1].

Why heavy nuclei need extra neutrons

Coulomb pushes $Z_{\min}$ below $A/2$. Each missing proton relative to $A/2$ is replaced by a neutron, so $N - Z = A - 2Z_{\min}$ grows with A . The asymmetry term fights that growth; equilibrium is the compromise in (1.23). Without Coulomb, $Z_{\min} = A/2$ exactly (up to the tiny $m_n - m_p$ shift).

1.8.2 Parabola accuracy and Q_β residuals

At fixed A , the SEMF reproduces mass *differences* between neighbouring isobars to within a few hundred keV for medium-mass nuclei once the overall offset is fixed by the lowest isobar. Shell corrections and pairing details appear as residuals of order 0.5–1 MeV [14, 6]. Near $Z_{\min}$ the Q_β values are therefore small; far up the parabola they grow, setting the scale for Fermi-theory spectra in a later chapter.

1.9 Physics depth: mass parabolas and $Z_{\min}$

At fixed A , the SEMF mass as a function of Z is a parabola (two parabolas when pairing splits even and odd Z). Minimising with respect to Z yields

$$Z_{\min} \simeq \frac{A}{2 + (a_C/(2a_a))A^{2/3}}, \quad (1.24)$$

which drifts below $A/2$ as Coulomb grows. That single formula explains why stable heavy nuclei are neutron rich.

1.9.1 Worked estimate: $A = 125$

With $a_C/(2a_a) \sim 0.015$, one finds $Z_{\min} \simeq 125/(2 + 0.015 \times 25) \simeq 52$, near tin: the valley of stability has already leaned toward $N > Z$. The curvature of the parabola sets Q_β for neighbours: each step in Z away from $Z_{\min}$ costs roughly $2a_a(A - 2Z)/A$ in asymmetry plus a Coulomb differential.

Odd- A vs even- A

Odd A has one parabola; even A has two, separated by $2|\delta|$. Double- β candidates sit where a nucleus is stable against single β decay but not against a second-order jump to the neighbouring even–even isobar. Lecture 19 returns to that geometry with Fermi theory.

Remark

Limits. Shell closures warp the parabola locally. Near magic numbers, $Z_{\min}$ from a global SEMF fit can be wrong by one unit, which is enough to mis-order Q_β signs for waiting-point nuclei.

1.9.2 Reading a mass parabola plot

Plot $M(A, Z)$ or the mass excess against Z at fixed A . The vertex is $Z_{\min}$; the local slope is proportional to $-Q_\beta$ for the adjacent decay. Mirror nuclei ($N \leftrightarrow Z$) in light systems test the Coulomb piece of the SEMF because the asymmetry term flips sign under that exchange. In medium mass, the valley's neutron excess is the cleanest classroom demonstration that Coulomb and asymmetry compete.

For even A , draw both pairing-shifted parabolas. A nucleus that sits in a local minimum on the even–even parabola but sees an odd–odd neighbour higher in energy may be stable against single β decay yet unstable against double β decay: that geometric fact is the starting point of $0\nu\beta\beta$ target selection.

For a concrete chain such as $A = 101$, compute $Z_{\min}$ from the analytic formula, then list Q_β for two steps on either side using SEMF masses. Compare with evaluated masses if available. The exercise trains the reflex that mass differences, not absolute masses, drive decay maps, and it shows how a 1 MeV shell correction can reorder two close isobars.

Worked $Z_{\min}$ drift

Plot $Z_{\min}(A)/A$ from the analytic formula for $A = 40$ to $A = 240$. The steady decline below $1/2$ is the Coulomb–asymmetry competition in one curve. Overlay a few evaluated stable Z values to see where shells pull the valley away from the smooth prediction.

Valley plot habit

Whenever you meet a new mass region, sketch $Z_{\min}(A)$ and two Q_β arrows. The habit prevents treating β stability as a memorised list of isotopes rather than a consequence of the energy surface from Lecture 4.

Physics-depth close

Mass parabolas turn the SEMF into a local decay map. Once $Z_{\min}$ and Q_β are in hand, spectroscopy and astrophysics share a common arithmetic front end.

1.10 Further physics: atomic minimum and chart systematics

1.10.1 Atomic mass minimum and the 0.78 MeV shift

Stability against β decay is decided by the *atomic* mass, not the bare nuclear mass. Minimising $M_{\text{atom}}(A, Z)$ includes the neutron–hydrogen mass difference $\delta_{nH} = (m_n - m(^1\text{H}))c^2 \approx 0.78 \text{ MeV}$ in $Z_{\min}$. Light nuclei slightly favour protons relative to a pure $N = Z$ Fermi-gas estimate, while Coulomb still forces $N > Z$ for heavy systems [14, 6].

1.10.2 Odd A one parabola; even A two

For odd A (pairing zero) masses follow one parabola in Z . For even A , even–even and odd–odd species form two parabolas split by $\sim 2|\delta|$. Double- β candidates sit on the even–even curve when single β would land on the higher odd–odd curve [14, 5, 13, 1].

A useful illustration is $A = 112$: ^{112}Sn and ^{112}Cd can both be stable against single β on the lower even–even parabola, while intermediate odd–odd ^{112}In lies higher and can decay both ways [14].

1.10.3 Systematics on the chart of nuclides

On the (N, Z) plane:

- nuclei *below* the valley (neutron excess) decay by β^- ;
- nuclei *above* the valley (proton excess) decay by β^+ or electron capture;
- the valley bends toward $N > Z$ as A grows, as $Z_{\min}(A) < A/2$.

Plotting the integer nearest to $Z_{\min}(A)$ reproduces the valley curve seen on the chart of nuclides in Lecture 4.

1.11 Deeper reading: from parabolas to decay maps

Once masses are known, $Q_{\beta^-} = M(A, Z) - M(A, Z + 1)$ (atomic convention) and electron-capture Q values follow by arithmetic. Precision Penning-trap masses revise those Q values at the kilo-electronvolt level, which matters for endpoint neutrino-mass experiments and for r -process pathfinding where photodisintegration thresholds sit near zero.

1.11.1 Bridge in the lecture arc

Lecture 4 supplied the smooth energy surface; Lecture 5 slices it at fixed A ; Lectures 18–19 turn the same Q into spectral endpoints and ft values. Neutron-star crust composition (Lecture 6) likewise tracks $Z_{\min}(A)$ under electron degeneracy pressure. A useful student habit is to compute both SEMF and AME-based Q values for one chain and to tabulate the residual: that residual is the microscopic correction the shell model must eventually explain.

1.11.2 Separation energies and drip lines

Neutron and proton separation energies S_n, S_p are discrete derivatives of the mass surface. The drip lines are where those derivatives change sign. Continuum coupling and pairing correlations shift the lines relative to a naive SEMF extrapolation, which is why rare-isotope mass campaigns are structure experiments as much as metrology. Lecture 13's single-particle picture and Lecture 16's decay Q values both inherit the same mass table.

For a concrete chain such as $A = 101$, compute $Z_{\min}$ from the analytic formula, then list Q_{β} for two steps on either side using SEMF masses. Compare with evaluated masses if available. The exercise trains the reflex that mass differences, not absolute masses, drive decay maps, and it shows how a 1 MeV shell correction can reorder two close isobars.

Parabola checklist

For one odd A and one even A , sketch the mass parabola(s), mark $Z_{\min}$, and compute two Q_{β} values from both SEMF and evaluated masses. Discuss any sign disagreement. That single exercise connects Lecture 4's energy surface to Lectures 18–19 decay spectroscopy and to drip-line forecasts used in Lectures 6 and 13.

Isobaric analogue and Coulomb displacement

Coulomb displacement energies between mirror states test the Coulomb term beyond the SEMF's smooth $Z(Z - 1)/A^{1/3}$ piece. Charge-symmetry breaking and nuclear-structure polarisation contribute at the hundred-keV level. For classroom purposes, comparing SEMF Coulomb differences with measured mirror mass differences in the sd shell already shows where the smooth drop formula is incomplete.

Waiting points

In the r -process, waiting-point nuclei have small S_n and long β lifetimes. A 100 keV mass error can move a photodisintegration threshold across a typical temperature window. Propagating

mass uncertainties through a network, rather than quoting only central $Z_{\min}$, is the modern standard foreshadowed by this chapter's parabola arithmetic.

Mass surfaces in three dimensions

It helps to visualise $M(A, Z)$ as a surface rather than a single parabola. Cuts at fixed A produce the Lecture 5 parabolas; cuts at fixed Z produce isotopic chains whose slopes are neutron separation energies; cuts at fixed N produce proton separation energies. Drip lines are the curves where those slopes change sign. Shell closures appear as local ridges. Once that three-dimensional picture is in mind, $Z_{\min}(A)$ is simply the valley floor's path through the chart.

For a quantitative exercise, download a small AME subset for $A = 100\text{--}110$, plot mass excess versus Z for each A , and mark the lowest point. Compare with the SEMF $Z_{\min}$. Where they disagree by one unit, look for a nearby magic number or deformation region. That disagreement is not a failure of pedagogy: it is the empirical signal that Lectures 12–15 must explain with shells and collectivity. Finally, convert two neighbouring masses into Q_β and estimate a rough allowed lifetime scale using the phase-space ideas of Lecture 19, even before a full Fermi calculation: the arithmetic chain from mass table to decay map should become automatic.

Mass parabolas turn the SEMF from a global fit into a local decay map. Once $Z_{\min}$ and Q_β are in hand, spectroscopy, astrophysics, and neutrino experiments share a common arithmetic front end.

In practice, rework one numerical estimate from the lecture with a different parameter set and compare the shift to the quoted experimental uncertainty. If the shift exceeds the uncertainty, the estimate is teaching a scale, not delivering a fit. That distinction keeps theoretical cartoons honest when they sit beside evaluated data tables and modern many-body calculations. Along the course arc, write one sentence linking this chapter backward and one forward: the discipline of those two sentences is as important as any single formula. Examination problems rarely ask for a literature review, but they do ask for the scale argument that literature reviews presuppose.

Research frontier: precision masses, $Z_{\min}$, and student projects

Lecture 5 connected $Z_{\min}(A)$ and mass parabolas to β -decay energetics: the most stable isobar for odd A sits at the bottom of a parabola, while even- A systems show pairing-split doublets. Today's frontier is Penning-trap and storage-ring mass spectrometry of short-lived nuclides, which maps Q_β and nucleon separation energies with keV (sometimes sub-keV) accuracy for nuclei far from stability [27, 51]. Those masses fix pathways in the r -process, determine whether a waiting point can be photodisintegrated in hot environments, and constrain neutron-star crust compositions when combined with multimessenger radius and tidal data [46]. Open issues include incomplete coverage along the most neutron-rich paths, isomer contamination in mass measurements, and the need to propagate mass uncertainties through network calculations rather than quoting only central values. Facility upgrades at FRIB, FAIR, and related laboratories are explicitly motivated by closing those mass gaps [52].

The connection back to the lecture is direct: every improved mass shifts $Z_{\min}$ and the Q values that set β endpoints. When a Penning-trap result revises a waiting-point mass by

tens of keV, network calculations can change both the abundance pattern and the freeze-out timescale. Learning to read AME uncertainty columns is therefore part of modern nuclear astrophysics literacy.

Student directions. (1) Data project: from AME2020, reconstruct the odd- A parabola for $A = 135$ and compare with the lecture figure. (2) Analytic estimate: compute Q_{β^-} along an isotopic chain and mark where S_n approaches zero. (3) Literature survey: propose a survey of one FRIB or FAIR mass-measurement campaign and state which nucleosynthesis observable it impacts.

1.12 Problems with solutions

Problem 5.1 — Evaluate $Z_{\min}$

Using $Z_{\min} = A/[2(1 + 0.0078 A^{2/3})]$, compute $Z_{\min}$ for $A = 100$ and $A = 200$. For each, give N/Z at that charge.

Solution 5.1

For $A = 100$, $A^{2/3} \approx 21.54$,

$$Z_{\min} = \frac{50}{1 + 0.0078 \times 21.54} = \frac{50}{1.168} \approx 42.8 \quad \Rightarrow \quad \frac{N}{Z} \approx \frac{57.2}{42.8} \approx 1.34.$$

For $A = 200$, $A^{2/3} \approx 34.2$,

$$Z_{\min} = \frac{100}{1 + 0.0078 \times 34.2} = \frac{100}{1.267} \approx 78.9 \quad \Rightarrow \quad \frac{N}{Z} \approx \frac{121}{79} \approx 1.53.$$

Heavy nuclei are neutron rich [6].

Problem 5.2 — Free-mass difference

Show that $(m_n - m(^1\text{H}))c^2 \approx 0.782 \text{ MeV}$ using $m_n = 1.008665 \text{ u}$, $m(^1\text{H}) = 1.007825 \text{ u}$, and $c^2 = 931.5 \text{ MeV/u}$. Where does this number enter $Z_{\min}$?

Solution 5.2

$$(1.008665 - 1.007825) \times 931.5 = 0.000840 \times 931.5 \approx 0.782 \text{ MeV}.$$

It appears in the numerator of the exact $Z_{\min}$ formula (1.13) as the free-mass drive toward protons; it is small compared with $4a_a \approx 92 \text{ MeV}$ [6, 14].

Problem 5.3 — Q_{β^-} sign on the parabola

At fixed odd A , $M(Z)$ is a parabola with minimum at $Z_{\min}$. Show that $Q_{\beta^-} = [M(Z) - M(Z + 1)]c^2 > 0$ for all integers $Z < Z_{\min} - 1/2$, so neutron-rich isobars β^- decay toward the minimum.

Solution 5.3

Write $M(Z) = \alpha - \beta Z + \gamma Z^2$ with $\gamma > 0$. Then

$$M(Z) - M(Z + 1) = \beta - \gamma(2Z + 1).$$

The minimum of M is at $Z_{\min} = \beta/(2\gamma)$, so

$$M(Z) - M(Z + 1) = 2\gamma(Z_{\min} - Z - \tfrac{1}{2}).$$

For $Z \leq Z_{\min} - 1$ (integer steps below the vertex), this difference is positive: $Q_{\beta^-} > 0$. Geometry alone forces β^- on the neutron-rich side [6, 1].

Problem 5.4 — Pairing split and double β

Even–even and odd–odd mass parabolas at fixed even A are split by $2|\delta|$ with $|\delta| = a_P A^{-3/4}$. For $A = 128$ and $a_P = 34$ MeV, estimate the vertical split $2|\delta|$. Explain in one sentence why an even–even nucleus can be stable against single β yet unstable against double β .

Solution 5.4

$$A^{3/4} = 128^{0.75} \approx (2^7)^{0.75} = 2^{5.25} \approx 38, \quad |\delta| \approx 34/38 \approx 0.9 \text{ MeV}, \quad 2|\delta| \approx 1.8 \text{ MeV}.$$

Single β would land on the higher odd–odd parabola, so $Q_{\beta} < 0$ for that step, while a jump of $\Delta Z = 2$ to a lower even–even neighbour can still have $Q_{2\beta} > 0$ [6, 1].

Problem 5.5 — Both β modes (Petrera-style)

For odd–odd ^{64}Cu near mid-mass, explain qualitatively why both β^- (to ^{64}Zn) and β^+ (to ^{64}Ni) can be open, using the two-parabola picture.

Solution 5.5

^{64}Cu sits on the upper (odd–odd) parabola. Both even–even neighbours ^{64}Ni and ^{64}Zn lie on the lower parabola and can be lower in mass than ^{64}Cu . Then both Q_{β^-} and Q_{β^+} are positive: bidirectional decay. Quantitative SEMF estimates give $Q \sim 1$ MeV for both branches [3, 6].

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